How Many Evaluator Scores Certify Output Retention Under a Churn Budget?

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Abstract

Replacing an evaluator can change which language outputs pass an ensemble cutoff. We study the score evidence needed to certify that a returned subset meets an average normalized churn budget and retains a count within a specified tolerance of the full-information optimum under that budget. The normalization, cutoff and a finite family of single-evaluator replacements are fixed, and the guarantees hold for every score completion consistent with the observed entries. We characterize minimum certificates chosen with full-score knowledge and verified from revealed entries only. For odd median pools, the evidence for feasibility and the evidence excluding larger retained sets are disjoint, which yields closed-form minima. Mean and even-median pools allow one observation to serve both purposes. Under exact count certification from an empty history, we construct mean-pooling families whose joint certificate approaches half the size of the best union of separately minimum certificates. Across 1,920 mean targets from cached RewardBench 2 and five WMT panels, numerical optimization finds modest joint gains in 89 targets, all of which require exclusion evidence, and the corresponding whole-column targets show no gain. Discovering a two-entry certificate can require queries linear in batch size, even with positive count tolerance. We also extend the analysis to weighted and per-replacement churn objectives and to verifiable cross-entry score constraints. Certificate size, discovery cost and inference time require separate evaluation, and retention stability does not certify output quality.

1 Introduction

Reward models, translation metrics and LLM judges score language outputs (Malik et al., 2026; Freitag et al., 2023; Zheng et al., 2023), and ensembles of such evaluators are often used to filter a batch (Coste et al., 2024; Eisenstein et al., 2024). Evaluators are replaced over time. Before rescoring a selected batch after replacing one ensemble member, an analyst may want evidence that many outputs would keep their selection status within a declared churn allowance. Returning nothing is always feasible, so a useful certificate must also bound how many outputs are lost relative to the best budget-feasible retention.

We study this problem for a fixed batch. A base pool of evaluators selects outputs whose median or mean normalized score reaches a cutoff, and a finite family of candidate evaluators defines every single-member replacement. Candidate scores on the batch are initially unknown and can be revealed entry by entry. A *certificate* is a set of revealed entries from which a verifier can conclude, for every completion of the unrevealed scores, that a returned subset meets the churn budget and that its count is within ε outputs of the optimum. This guarantee is deterministic and completion-wise. It complements conformal and selective risk control, which bound expected risk under exchangeability (Angelopoulos et al., 2024; Bai and Jin, 2026).

Our contributions are as follows. (i) We define safety evidence, which certifies feasibility, and exclusion evidence, which rules out larger retained sets, and we show that both bounds are sharp under the declared completion model (Section 4). (ii) For odd medians the two kinds of evidence are disjoint, giving closed-form minimum certificates. For mean and even-median pools a single entry can serve both, and we give an exact joint program together with a tight factor-two separation from the best union of separately minimum witnesses (Section 5). (iii) We separate certificate size from discovery: two-entry certificates can require queries linear in batch size even at positive tolerance (Section 6). (iv) On RewardBench 2, WMT and SummEval panels we measure inspection, minimum evidence, and joint gains, and we identify when sharing evidence helps (Section 7). (v) We extend the theory to weighted and per-replacement churn

objectives and to verifiable cross-entry constraints (Appendices R–S). Stability differs from quality: evaluator agreement can coexist with shared errors and reward hacking (Eisenstein et al., 2024).

2 Related Work

Risk control and judge stability. Conformal risk control (CRC) bounds an expected monotone loss (Angelopoulos et al., 2024; Yeh et al., 2025), and risk-controlling prediction sets give high-probability bounds (Bates et al., 2021). SCoRE, SCOPE and multi-expert CRC address selective risk, pairwise acceptance and multiple judges (Bai and Jin, 2026; Badshah et al., 2026; Cheng et al., 2026). Replicable calibration stabilizes thresholds across samples (Papamichalis et al., 2026). LLM judges are sensitive to position and prompt variations (Wang et al., 2024; Thakur et al., 2024), and ensemble agreement need not exclude common errors (Afrin and Shihab, 2026). Predictive churn has been studied for model updates and training-sample changes (Milani Fard et al., 2016; Watson-Daniels et al., 2025; Prastalo and Jablonka, 2026), and fixed-threshold routing can change even with matching class predictions (Kim, 2026). Our guarantee instead concerns one fixed batch, holds for every consistent score completion, and is stated relative to a declared replacement family.

Verification, discovery and robust selection. Priced information and matroid verification distinguish proof size from discovery cost (Charikar et al., 2002; Diwan et al., 2026), and Boolean evaluation studies adaptive queries (Deshpande et al., 2016; Gkenosis et al., 2018). Robust selection and knapsack problems choose queries before uncertain parameters are resolved (Chen et al., 2026; Goerigk et al., 2015), often with budgeted uncertainty sets (Bertsimas and Sim, 2004); uncertain-profit knapsack certifies good packings and excludes better ones under resource augmentation (Schlöter, 2025). Recoverable selection permits later exchanges (Chassein et al., 2018), and covering and query-competitive models study sharing and discovery (Megow and Schlöter, 2023; Erlebach and Hoffmann, 2015; Erlebach et al., 2021). These models leave costs or profits uncertain. Here the uncertainty is a matrix of candidate scores under a known pooling rule, and we verify feasibility and optimality for a fixed scenario family rather than solve a min–max design problem. Unlike black-box certification (Blanc et al., 2022; Gupta and Manoj, 2023), we query a fixed matrix.

Top-k with partial information. Sorted-list aggregation supports sorted and random accesses (Fagin et al., 2003); our certificates assume no sorted oracle. Top-k queries over uncertain or incomplete data compute possible and certain answers (Imieliński and Lipski, 1984; Soliman et al., 2007; Amarilli et al., 2017), and partial voting profiles admit necessary and possible winners (Imber and Kimelfeld, 2020). Our safe and optimistic counts play the role of certain and possible retention, but the object certified is a budget-constrained optimum over a replacement family, which leads to different certificate combinatorics. Ranking robustness covers weight changes and word attacks (Asudeh et al., 2018; Wu et al., 2022); PAC selection, Col-Bandit and budget-adaptive ranking acquire scores adaptively (Kalyanakrishnan et al., 2012; Pony et al., 2026; Oettershagen, 2026). Selective ensembles, pruning and early exit reduce scoring cost (Black et al., 2022; Li et al., 2022; Inoue, 2019; Busolin et al., 2021), and Best-of-Evidence queries factors within a fixed pool (Zhang et al., 2026). MT filtering and metric biases motivate separate quality assessment (Perrella et al., 2024; Shayegh et al., 2025).

3 Selection under Evaluator Replacements

Rows index outputs and columns index evaluators. Pools P and R contain $M \geq 1$ base and $T \geq 1$ candidate evaluators. Normalization maps and a cutoff $0 < t_0 \leq 1$ are fitted on a reference split of scored outputs (here, *reference* denotes the fitting split, not a human reference translation). Write $s_e(i) \in [0, 1]$ for normalized scores. Replacing base member j by candidate r gives the pool $P^{j \to r}$; the $H = MT$ replacements have uniform weight. The maps and cutoff remain fixed. Base scores are visible; queries reveal candidate test scores.

The base list is $S_0(P) = \{i : c_P(i) \geq t_0\}$, where c_P is median or mean consensus; an even-sized median averages the two central scores, and cutoff ties are included. For a batch of N outputs, fix a nominal selection fraction $q \in (0, 1]$ and the nominal count $k = \max(1, \mathrm{round}(qN))$ (half-integers round to even). The count k scales the risk and imposes no return quota. With $\sigma_Q(i) = \mathbf{1}[i \in S_0(Q)]$, base scores and one candidate score determine all M replacement outcomes of a row.

Quantity	Definition	Example
Churn budget	$R(C) \le \alpha$, Eq. (2)	.10
Changes/replacement	$\rho = 2k\alpha$	0.4
Flip budget	$b_H = \lfloor \rho H \rfloor$	8
Oppositions (odd M)	$b = \lfloor b_H/(h+1) \rfloor$	2
Count tolerance	$V^+ - V^- \le \varepsilon$	$\varepsilon \ge 0$
Active inspection	$100\,Q/(mT)$	–

Table 1: Units and conversions. The example uses $M = 7$ ($h = 3$), $T = 3$ and $k = 2$, so $H = 21$ (Appendix A.7). Because $2k$ is a fixed nominal scale, $R(C) > 1$ is possible when many outputs are returned; only $2k\alpha$ matters for feasibility (Appendix L).

Define the flip count and row total

$$f_{ir} = \sum_{j=1}^{M} \mathbf{1}[\sigma_P(i) \neq \sigma_{P^{j\to r}}(i)], \quad F_i = \sum_{r=1}^{T} f_{ir}. \tag{1}$$

A commitment mask C fixes baseline decisions without review and returns $S_0(P) \cap C$. Its average normalized churn over the declared replacements is

$$R(C) = \frac{1}{H} \sum_{j,r} \frac{|(S_0(P) \triangle S_0(P^{j\to r})) \cap C|}{2k} = \frac{\sum_{i\in C} F_i}{2kH}. \tag{2}$$

Decisions outside C are referred for review. We maximize the returned count subject to $R(C) \le \alpha$. Nonselected outputs do not increase the returned count and receive no dedicated entry queries, and an exclusion is committed only when base scores rule out every membership flip. With all costs observed, the integer flip budget is $b_H = \lfloor 2k\alpha H \rfloor$; sorting selected outputs by increasing F_i and taking the longest prefix within b_H gives the full-information optimum V^*.

Units. Table 1 collects the conversions. Allowing ρ average membership changes per replacement gives $\alpha = \rho/(2k)$. This is an average replacement allowance, not a per-replacement limit or a review-capacity quota.

Choice of risk objective. The uniform average in Eq. (2) makes the full-information problem a sorted prefix and yields the closed forms below. Three alternatives are natural. *Fixed nonnegative replacement weights* preserve every bound and the sorted-prefix argument; Theorem 3 extends with column-dependent caps, and odd-median evidence remains disjoint (Appendix R.1). *Certifying each single replacement*, i.e., a per-replacement cap $\max_{j,r} |\cdot| / (2k) \le \alpha$, keeps verification of a given mask linear-time and sharp, but computing the optimal retained count becomes NP-hard even with full information, so count gaps require integer programs (Appendix R.2). This matters in practice: 6,985 of 15,360 average-certified masks exceed .10 for at least one individual replacement (Appendix I). *A known single replacement* reduces to one candidate column with Boolean row costs. Uniform weights correspond to a replacement chosen uniformly from the declared family.

4 Certificates from Partial Scores

A certificate uses visible base scores and queried candidate scores to verify churn feasibility and bound the count shortfall. One entry query determines all M replacement outcomes of its row. Let $f_i(x)$ count flips when a candidate assigns score x to output i, so $f_{ir} = f_i(s_r(i))$ and $p_i = \max_{x\in[0,1]} f_i(x)$. Under either pooling rule, replacement consensus is nondecreasing in x; for selected outputs f_i is nonincreasing, with maximum p_i at $x = 0$ and zero at $x = 1$. Visible base scores determine p_i. After t_i queries on row i, let ℓ_i sum their observed contributions. Bounding each unobserved count between zero and p_i gives

$$\ell_i \le F_i \le u_i = \ell_i + (T - t_i)p_i. \tag{3}$$

For nonnegative costs w, define the retention capacity $V(w) = \max\{|C \cap S_0(P)| : \sum_{i\in C} w_i \le b_H\}$, computed by sorting. Maximizing retention under u yields a mask returning $V^- = V(u)$ outputs that is feasible for every consistent completion; the optimistic count $V^+ = V(\ell)$ is attained in some completion.

The completion model. Unobserved scores range freely over $[0, 1]$. This is a set of possible matrices, not a statistical assumption: assigning every unobserved entry its zero-flip endpoint realizes ℓ, and assigning its maximum-flip endpoint realizes u. We use this worst-case set because correlations among evaluators, however strong on reference data, do not by themselves guarantee where an unseen test score lies; predictions therefore guide query order but never tighten bounds. Valid cross-entry constraints can only shrink the completion set: every certificate in this paper remains valid under them and minimum certificates can only become smaller. A verifiable class stays tractable. If the

constrained completion set is a lattice, with coordinatewise greatest and least elements, then the same two completions give sharp bounds. This covers per-entry score supports, monotonicity constraints and Lipschitz-with-slack constraints across outputs for each evaluator, whose envelopes are computed in linear time per query (Propositions 18–19, Appendix S). In a controlled study with valid Lipschitz constraints, certification inspects 3–6% of active entries instead of 65–80%; with a misspecified constant the guarantee is lost, so such constraints must be justified externally.

Proposition 1 (Safe retention and stopping)**.** *At any adaptive query history, let C^- maximize retained count subject to the upper-cost budget. Then $R(C^-) \le \alpha$ and $V^- \le V^* \le V^+$. If $V^+ - V^- \le \varepsilon$, its returned count is at most ε outputs below V^*. Under the declared replacement family and completion model, these guarantees hold for every acquisition order and stopping time.*

Proof. Equation (3) bounds each true cost. Upper-cost feasibility implies true feasibility, which implies lower-cost feasibility; maximizing cardinality gives the count bounds. The returned count is V^-, so its shortfall is at most the gap. □

Integer $\varepsilon \ge 0$ bounds count shortfall, and zero requires count optimality. Full revelation gives $V^- = V^+ = V^*$. An output is structurally invariant ($p_i = 0$) if no candidate score can change its membership; noninvariant outputs can still have zero true cost. Zero-query retention uses upper costs Tp_i and can admit noninvariant outputs within budget. Let $m_0 = |\{i \in S_0 : p_i = 0\}|$ count invariant selected rows; only the $m = |\{i \in S_0 : p_i > 0\}|$ active selected rows are eligible for entry queries, and active inspection is $Q/(mT)$ for Q entry queries (zero when $m = 0$). The safe count cannot decrease, although retained sets need not be nested.

5 Minimum Certificate Complexity

Odd medians: disjoint evidence. For $M = 2h + 1$, median selection requires $h + 1$ votes at or above t_0, and a single replacement changes at most one vote. Selected rows with at least $h + 2$ votes are invariant; those with exactly $h + 1$ form the boundary set B, and a below-cutoff candidate excludes a boundary row exactly when it replaces one of its $h + 1$ voting members. Thus $p_i = h + 1$ on B. For $i \in B$ define opposition bits $y_{ir} = \mathbf{1}[s_r(i) < t_0]$ and row sums $z_i = \sum_r y_{ir}$, so $F_i = (h + 1)z_i$. In opposition units, feasibility is exactly $\sum_{i \in C \cap B} z_i \le b$ with $b = \lfloor b_H/(h + 1) \rfloor$ (Table 1). On a boundary row, an *agreement* (candidate score at or above the cutoff) lowers the upper bound by one opposition, and an *opposition* raises the lower bound by one: with a_i agreements and o_i oppositions observed, $o_i \le z_i \le T - a_i$. An offline constructor uses the true bit matrix to choose entries; the verifier sees only revealed entries and checks every consistent Boolean completion.

Theorem 1 (Safety witness)**.** *If the true matrix admits a feasible set of L boundary rows, the minimum number of observations certifying that an L-row set is feasible under every consistent completion is $Q_{\text{safe}}(L) = \max\{0, LT - b\}$. If no L-row set is truly feasible, no certificate exists.*

An L-row set starts with upper cost LT, and only agreements reduce it, one unit each; any truly feasible set contains enough agreements (Appendix A gives proofs and a worked example). Feasibility need not establish optimality. Let $z_{(1)} \le \cdots \le z_{(m)}$ and let L be the largest prefix size with sum at most b; the full retained optimum is $m_0 + L$.

Theorem 2 (Exact optimality witness)**.** *If $L = m$, the minimum certificate proving exact optimality is $\max(0, mT - b)$. Otherwise set $p = L + 1$, $B_0 = b + 1$, $D(t) = \sum_{i=1}^{p} \min(z_{(i)}, t)$ and $t^* = \min\{t \in \{0, \ldots, T\} : D(t) \ge B_0\}$. The exact offline certificate size is*

$$Q_{\text{exact}} = \max(0, LT - b) + B_0 + (m - p)t^*. \quad (4)$$

The exclusion part reveals $b + 1$ oppositions across the p cheapest rows, capped at $\min(z_{(i)}, t^*)$, and t^* oppositions in every remaining row. These cells are disjoint from the safety agreements, so the two costs add. With certified safe boundary count s, tolerance ε requires every $(s+\varepsilon+1)$-row boundary set to be excluded unless that size exceeds m.

Corollary 1 (Minimum gap witness)**.** *Let L^* be the true optimal boundary count and $\varepsilon \ge 0$ an integer. For $L^* < p \le m$, let $D_p(t) = \sum_{i=1}^{p} \min(z_{(i)}, t)$, $t_p = \min\{t : D_p(t) > b\}$ and $E(p) = b + 1 + (m - p)t_p$; for $p > m$ set $E(p) = 0$. Then*

$$Q_\varepsilon = \min_{s \in \mathcal{S}_\varepsilon} \big[\max(0, sT - b) + E(s+\varepsilon+1) \big], \quad (5)$$

with $\mathcal{S}_\varepsilon = \{\max(0, L^* - \varepsilon), \ldots, L^*\}$. In particular $Q_0 = Q_{\text{exact}}$.

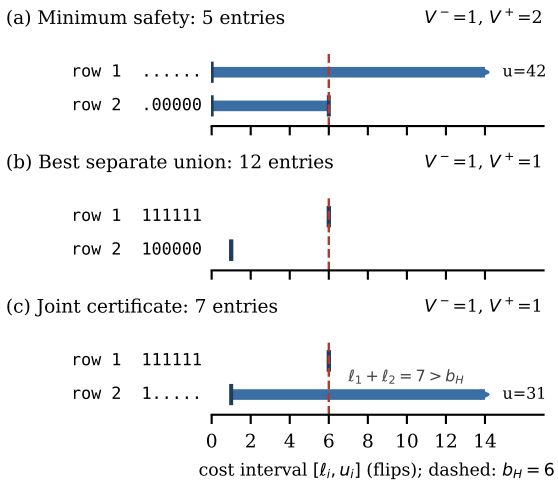


Figure 1: One mean-pooling entry can serve safety and exclusion (Proposition 9 at $T = 6$; caps $p = (7, 6)$, $b_H = 6$, exact stopping). Digits are revealed flip counts; dots are unqueried. Bars show $[\ell_i, u_i]$. (a) The shortest safety proof certifies row 2. (b) Its best union with the unique minimum exclusion witness reveals all 12 entries. (c) Retaining row 1 instead uses the six entries that exclusion already needs: they set $u_1 = 6 \le b_H$ (safety) and, with one entry of row 2, make the two smallest lower costs sum to $7 \ge b_H + 1$ (exclusion).

These minima lower-bound entry queries for the same instance, budget, model and tolerance from an empty history (Appendix A.6); residual witnesses account for existing observations (Appendix M.4).

Joint evidence: mean and even medians. An entry with true contribution f_{ir} raises its row's lower cost by f_{ir} and reduces its upper cost by $p_i - f_{ir}$. Mean pooling can have integer contributions $0 < f_{ir} < p_i$, so one query contributes to both proofs. Figure 1 shows the mechanism. Exclusion of every p-row set is equivalent to the p smallest lower costs summing to at least $b_H + 1$, which we write with a single threshold t as $\sum_i (t - \ell_i)_+ \le pt - b_H - 1$ (Lemma 1, Appendix M.5). In the figure, the six entries of row 1 are needed for exclusion; choosing row 1 as the retained row turns them into a safety proof as well, so the longer safety proof yields the shorter complete certificate.

For a history $\mathcal{O}$, let ℓ_i^0, u_i^0 be its row bounds and U_i its unqueried candidates. Using true contributions offline, define $\mathcal{D}_i = \{(|J|, \sum_{r\in J} f_{ir}) : J \subseteq U_i\}$. A state (a_i, v_i) gives $\ell_i = \ell_i^0 + v_i$ and $u_i = u_i^0 - a_i p_i + v_i$. Let L^* be the true optimal active count.

Theorem 3 (Joint entry witness)**.** *For the fixed true matrix, history $\mathcal{O}$ and integer $\varepsilon \ge 0$, the minimum additional entry count Q_{joint} is $\min \sum_i a_i$ over $s \in \{\max(0, L^* - \varepsilon), \dots, L^*\}$, row states in $\mathcal{D}_i$ and a safe set C of s active rows with $\sum_{i\in C} u_i \le b_H$. If $p = s + \varepsilon + 1 \le m$, also require an integer $0 \le t \le TM$ such that $\sum_i (t - \ell_i)_+ \le pt - b_H - 1$, where $(x)_+ = \max(0, x)$. For $p > m$, exclusion is automatic.*

Every successful query set induces reachable row states and a safe count, and every feasible state choice is attained by a query set of the same size (Appendix M.5). The offline dynamic program is polynomial in explicit m, T, M but has a budget-indexed grid; an equivalent integer program groups equal within-row outcomes into G row–outcome groups with $O(G+m)$ variables and constraints per safe-count target, without guaranteeing polynomial solver time. On the 1,920 full mean targets, this program closes 1,876 within a nominal 60 seconds, and constructed-score grids up to $N = 8192$, $T = 65$ close in four of eight cases (Appendix M.7). Its optima are identical across HiGHS settings and CBC on our solver-sensitivity grid (Appendix W).

What joint optimization adds. Let Q_{split} minimize the sum of separately minimum safety and exclusion sizes over admissible safe counts, and let $Q_{\cup,\text{sep}}$ minimize the union of tied minimum safety and exclusion witnesses at the same count, then over counts. For the same instance, history and tolerance,

$$Q_{\text{joint}} \le Q_{\cup,\text{sep}} \le Q_{\text{split}} \le 2Q_{\text{joint}}. \qquad (6)$$

Realizable mean families attain $Q_{\cup,\text{sep}}/Q_{\text{joint}} \to 2$, or $53/30$ at fixed $M = 7$, $\alpha = .10$, with empty history and exact stopping (Appendix M.8). The sandwich holds for any fixed nonnegative replacement weights, and the factor-two family persists for every weight vector within relative distance $1/(2MT)$ of uniform (Proposition 14, Appendix R.1).

When can sharing help? Three conditions are checkable before any candidate-test score is seen. A strict gain $Q_{\cup,\text{sep}} > Q_{\text{joint}}$ requires (a) a nonzero initial gap, $|S_0| - V^-(0) > \varepsilon$, and (b) an active row with two distinct replacement thresholds in $(0, 1]$, since otherwise every contribution is 0 or p_i and Corollary 3 gives equality; odd medians never satisfy (b). Ex post, (c) a gain requires a witness that cannot certify exclusion automatically, and all 1,831 no-gain NLP targets have a witness with $V^- + \varepsilon \ge |S_0|$ (Appendix M.9). A reference-predicted version of (c), $\hat{V}^* + \varepsilon < |S_0|$, is a usable

pre-scoring screen (Appendix U). A history alone can never certify a strictly positive remaining advantage, because a completion with zero unseen contributions makes safety alone sufficient (Proposition 6).

6 Acquiring a Certificate

Fixed order uses seeded row priority followed by candidate index; random order permutes active entries; largest width maximizes $u_i - \ell_i$ over eligible rows. Reference priority fits one ridge model per candidate to reference-row flip counts f_{ir} (features: pool scores, cutoff margin, score range and p_i/M; predictions clipped to $[0, p_i]$, Appendix B). It selects the eligible row with the smallest estimated total cost and, within it, the unqueried candidate with the largest predicted upper-cost reduction $p_i - \hat{f}_{ir}$. Frontier priority (mean pooling) clips predicted active capacity to $[V^- - m_0 + 1, V^+ - m_0]$, obtaining a target s, and scores predicted progress towards safety for s and exclusion of $s + 1$ (Appendix P.3). For odd medians, the closed forms suggest a targeted policy, OM-CF, that pursues agreements in the predicted safe set and capped oppositions at the Theorem 2 level (Appendix T). Predictions guide order, not bounds.

These policies differ in what they can guarantee. With exact predictions, OM-CF stops after exactly Q_{exact} queries (Proposition 20), so the closed forms make a prediction-based policy consistent in the sense of learning-augmented algorithms (Lykouris and Vassilvitskii, 2018; Purohit et al., 2018). No policy, however, can be robust: Propositions 7 and 8 give instances with fixed $M = T = 7$, $q = 1/20$, $\alpha = 1/10$ where any always-valid policy needs queries linear in batch size, even at the primary positive tolerance, while the minimum certificate has one or two entries. Sorting computes each capacity bound in $O(m \log m)$ time.

7 NLP Score-Inspection Experiments

7.1 Panels and protocol

RewardBench 2 has 1,723 four-completion prompts and 14 reward models (Malik et al., 2026). WMT23 en-de has 557 source segments, 13 output streams and 24 reference-based metrics (Freitag et al., 2023); removing `synthetic_ref` gives a twelve-stream variant with refitted maps and cutoff. The frozen first-seven pools leave $T = 7$ RewardBench or $T = 17$ WMT23 candidates (Appendix B); Section 7.4 tests composition sensitivity. Forty seeded splits preserve prompt/source groups. Each evaluator uses a reference-fitted right empirical cumulative distribution function (ECDF), and the cutoff is the $\max(1, \text{round}(qn_{\text{ref}}))$-th highest reference consensus. We use both pooling rules, $q \in \{.05, .10, .20, .30\}$ and $\alpha \in \{.02, .05, .10, .15, .20\}$ (primary .10). Primary stopping uses $V^+ - V^- \le \varepsilon = \lceil .02|S_0(P)| \rceil$, returning $K = V^-$ so that $V^* - K \le \varepsilon$; exact stopping uses zero. Queries access cached scores; reference and base scoring are charged separately, and acquisition counts exclude panel construction and full-panel WMT23 preprocessing reads. These are logical access counts, not inference savings, and the overlapping splits support descriptive summaries.

7.2 Retention baselines and risk guarantees

Selected coverage is $K/|S_0|$. At $q = .05$, $\alpha = .10$ and median pooling, zero-query coverage spans 80.9–99.2% across the three original panels, versus 99.5–99.9% under full information (Appendix Table 8); invariant and budget-admissible outputs both contribute (Appendix G.1). Fixed-count, batch-feasible and calibrated baselines have different constraints and guarantees (Appendix C). Envelope CRC calibrates an output-cost threshold with the loss-envelope correction of Yeh et al. (2025); under group exchangeability its expectation guarantee does not ensure the budget on every batch. Grouped SCoRE controls marginal deployment risk through whole-group decisions; at $q = .05$, $\alpha = .10$, RewardBench coverage is 96.4% (median) and 98.8% (mean) without candidate test scoring for selection, but unequal-size WMT24 documents are excluded. Tables 26–31 compare returned counts and churn with reference, calibration and test costs charged in the same units; the guarantees remain distinct.

7.3 Inspection and certificate size

At $q = .05$ (Table 2), reference priority averages 61.4% active-entry inspection with median pooling on RewardBench and 45.1% on thirteen-stream WMT23; mean pooling averages 85.4–86.5% across the three panels. Twelve-stream WMT23 under median pooling stops at zero on 37/40 splits.

Matching minima. Comparing exact odd-median stopping with its matching minimum (Theorem 2), the median splitwise reference-

Panel	Pooling	Z (%)	Fixed (%)	Random (%)	Width (%)	Ref. (%)	Cov. (%)
RewardBench 2	median	0.0	63.8	64.0	63.0	61.4	97.3
WMT23, 13 streams	median	10.0	57.9	58.4	65.0	45.1	97.7
WMT23, 12 streams	median	92.5	3.9	3.8	3.9	3.8	99.6
RewardBench 2	mean	0.0	90.9	91.5	91.7	86.5	97.5
WMT23, 13 streams	mean	0.0	91.3	92.2	92.6	85.7	97.5
WMT23, 12 streams	mean	0.0	89.5	90.2	89.9	85.4	97.0

Table 2: $q = .05$, $\alpha = .10$, gap $V^+ - V^- \leq \lceil .02|S_0| \rceil$. Z: percentage of 40 splits stopping without queries. Order columns: mean active inspection ($100Q/(mT)$), zero for empty active sets. Ref.: reference priority; Cov.: its mean selected coverage ($100K/|S_0|$).

query/Q_{exact} ratio is 1.968 for RewardBench and 1.662 for thirteen-stream WMT23, conditional on $Q_{\text{exact}} > 0$ (Appendix Table 36, M.2). Positive-gap comparisons use Q_ε (Appendix N). RewardBench 2, two WMT23 variants and three WMT24 directions give six panels; four depths and 40 splits yield 960 mean instances and 1,920 targets across the two stops. Each joint and optimal-union witness matches a numerical global lower bound (Appendix M.9; Table 3); this is distinct from an exact rational proof certificate.

Beyond deduplication: when do gains occur? Joint optimization improves on $Q_{\cup,\text{sep}}$ in 89 of the 1,920 targets, saving 6,269 entries in total (Appendix M.9). Median savings are zero; maximum reductions relative to $Q_{\cup,\text{sep}}$ are 2.61% (primary) and .87% (exact), with mean savings of 5.34 and 1.19 entries (Table 56). The gains concentrate where exclusion evidence is necessary. Full retention is feasible in 903 of the 960 instances, which makes exclusion automatic in their 1,806 targets. Of the 57 instances that cannot retain the full baseline list, all 57 exact targets and 32 primary targets show strict gains, and the remaining 25 primary targets have an optimal witness that certifies at least $|S_0| - \varepsilon$ outputs safe (Table 57). WMT24 ja-zh contributes 63 of the 89 gains. The prevalence of intermediate contributions $0 < f_{ir} < p_i$ does not explain gains on its own: in 13 strata containing both outcomes, gain targets have the larger intermediate fraction in eight and the smaller in five. Multiplying the budget by .75 and 1.25 moves the number of non-full instances from 57 to 174 and 19 (Table 60), so tighter budgets enlarge the region where sharing can matter. All 8,640 whole-column tasks on this grid have equal joint and best separate-union minima.

Online discovery. Across the 960 primary mean targets, frontier priority averages 5210.6 queries versus 5286.5 for reference and 5349.7 for width. It beats reference in 403 instances and loses in 554, with three ties. The lower mean despite more losses has a simple diagnosis: the exclusion term helps when exclusion is necessary (it saves 20.28 entries on average over safety-only on the 32 such targets) and costs a few entries elsewhere (3.72 on the 903 full-retention targets), so frontier's wins are larger than its losses (Appendix P.3, Table 71). Safety-only frontier has a slightly lower overall mean, and the ablations do not establish that both terms are needed.

7.4 Budget and evaluator composition

Inspection need not decrease with the churn budget: safety can become easier while exclusion becomes harder (Appendix G). Composition changes both the replacement family and the selected set; Appendix G reports twenty random pools per original panel and WMT23 pools with a fixed mix of surface-overlap and other metrics.

7.5 External panels and quality diagnostics

WMT24 en-de, en-es and ja-zh use document splits, exclude human-reference outputs and retain four depths, 40 splits and $\alpha = .10$ (Freitag et al., 2024). At $q = .05$, median zero-query coverage is 98.1–99.8%. The objective and panel criteria preceded WMT24 result inspection, but numerical priorities were repaired after a replay failure (a partial freeze; Appendix H). Quality is assessed after selection. RewardBench precision is the preferred-label fraction among returned outputs; its mean differences from label-blind, same-consensus top-K are -0.46 to -0.03 percentage points across depths under median reference priority. Released MQM differences have either sign; missing labels are not imputed (Appendices E and H).

7.6 Reference mapping and query granularity

On thirteen-stream WMT23 with median pooling at $q = .05$, switching from full-reference right to

Panel	Num. closed	Primary Q	Primary Ref./Q	Exact Ref./Q	Gain/num. closed
RewardBench 2	320/320	1216.9	1.205	1.186	0/320
WMT23, 13	320/320	2012.1	1.394	1.318	7/320
WMT23, 12	320/320	2024.3	1.291	1.258	13/320
WMT24 en-de	320/320	6922.7	1.185	1.169	3/320
WMT24 en-es	320/320	5968.4	1.187	1.174	3/320
WMT24 ja-zh	320/320	8717.0	1.119	1.107	63/320

Table 3: Full-test mean evidence at $\alpha = .10$: four depths, 40 splits and two count-gap targets per panel. Primary uses $\varepsilon = \lceil .02|S_0| \rceil$; exact uses zero. Primary Q averages the minimum joint entry count over 160 primary targets per panel; each ratio uses its matching target. All joint and union minima are numerically closed. Ratios average splitwise reference queries divided by positive matching minima. Gain compares joint evidence with the best union of separately optimal witnesses. Appendix M.9 reports all depths.

midrank ECDF raises zero-query selected coverage from 84.9% to 99.8% and lowers mean active inspection at primary stopping from 45.1% to 0.3%. Test groups stay fixed, but baseline-set Jaccard similarity is 0.77: midrank reduces the share of selected outputs exactly at the cutoff from 28.8% to 7.3%, so the saving concerns a changed selection task (Appendix K). Difficulty is therefore not an intrinsic property of the data. We recommend fixing the mapping, reference-size rule and tie convention as part of the selection protocol before any candidate-test score is seen, choosing them for selection validity (e.g., the score resolution of each metric, Deutsch et al. 2023) rather than to minimize queries, and reporting reference-bootstrap sensitivity (Appendix K.2) together with the share of selected outputs at the cutoff. Replicable calibration can stabilize the cutoff, but not a difficulty that depends on the test batch (Appendix V).

At $q = .05$ and $\alpha = .10$, median RewardBench reference priority reveals 1.77% of all NT candidate-test entries through entry queries versus 71.8% through columns, which charge all N entries per column. Every column is required on all 114 non-full mean targets; a sufficient range condition covers 101 (Appendix M.12), while other realizable instances permit strict column gains.

7.7 Summarization and real scoring

A frozen SummEval protocol (Fabbri et al., 2021) tests all ten three-base/two-candidate compositions. Under mean pooling, joint frontier saves two queries on average when $V^* + \varepsilon < |S_0|$ but adds queries in full-retention cases. On a fresh five-metric panel, 192 A40 runs cover eight methods, four depths, two pools and three repeats; primary reference priority averages 89.97 s versus 108.18 s for full scoring, including common setup (Appendix Q).

7.8 Discussion and practical guidance

Joint optimization couples retention to evidence, extending the verification–discovery distinction (Charikar et al., 2002; Diwan et al., 2026) to a setting where the retained set is also chosen. Its empirical gains are modest, and discovery is the larger cost: on the same 960 primary targets, reference queries exceed Q_{joint} by 809.58 entries on average, whereas $Q_{\cup,\text{sep}} - Q_{\text{joint}}$ averages 5.34. Even true-score priorities fail to yield query optimality (Appendix P.5). For practitioners we suggest: (1) fix the replacement family, weights, mapping and cutoff before scoring, and choose the average, weighted or per-replacement objective to match how replacements will be executed; (2) check the initial gap $|S_0| - V^-(0)$ first, since it decides whether any query is needed; (3) invoke joint optimization only when exclusion is likely necessary ($\hat{V}^* + \varepsilon < |S_0|$), otherwise a minimum safety witness suffices; (4) report evidence size, queries and scoring time separately, together with externally labeled quality assessments.

8 Conclusion

Retention evidence depends on safety, exclusion and query granularity. Odd medians separate the two proof tasks; mean pools permit sharing beyond optimal separate unions, with modest empirical gains confined to targets that need exclusion evidence. The analysis extends to weighted and per-replacement objectives and to lattice-structured score constraints. Small certificates ensure neither inexpensive discovery nor inference.

Limitations

The main certificates cover uniformly weighted single-member replacements from the declared finite family, with fixed maps and cutoff. They con-

trol average churn without requiring each replacement to satisfy α and do not cover evaluators outside the family; drift beyond the family requires extending the family and its budget. Per-replacement caps make the full-information optimum NP-hard (Appendix R.2), and weighted odd-median minima no longer have closed forms. Adding original-batch outputs to the returned set requires rechecking its budget, and newly generated outputs change the batch being certified. Nominal $2k$ normalization does not constrain the returned count to k. The odd-median closed forms assume unit-cost entry queries and Boolean completions without cross-entry constraints. Constrained completions (Appendix S) are valid only if the constraints hold for the true scores; learned correlations do not establish this, and minimum-certificate formulas do not transfer to them. The near-full criterion (Proposition 11) assumes empty history, exact stopping and a restricted family of minimum exclusion witnesses. The joint dynamic program has a large count-indexed state space, and outcome grouping reduces model size without guaranteeing polynomial solver time. Forty splits overlap, so aggregate comparisons are descriptive. True-score and oracle priorities are diagnostics, not deployable policies, and OM-CF with learned predictions did not improve on reference priority in our controlled study. The revision experiments in Appendices R–W use controlled synthetic panels and were not rerun on the NLP panels.

RewardBench and WMT23 were used during method development. WMT24 results had not been inspected when the risk objective and panel-inclusion criteria were fixed; after an exact-replay failure, numerical query priorities were corrected, which affected query order but not the certificate definition. This partial freeze is not a fully preregistered evaluation. Independence between evaluator training data and the benchmarks is not established, and some released meta-metrics use human-score calibration. The rationale for the original frozen evaluator ordering is not established by the records, and sampled pools do not cover all future evaluators. Reference priority has no online competitive guarantee. Incomplete MQM coverage restricts quality diagnostics to annotated outputs. The SummEval timing study measures five fixed scorers on one A40 with batch-eight full scoring and sequential entry queries; its three repetitions measure runtime variability rather than independent task replication.

Ethical Considerations

The experiments reuse evaluator panels and existing labels or generate controlled numerical scores; they collect no new human labels or language responses. Source reconstruction matches existing benchmark text to cached scores and output identities locally. Available records do not document a systematic audit for personally identifiable information or offensive content. Stable selection can preserve evaluator errors, so a certificate does not endorse retained content. Appendices B and H document data sources and acquisition. We have not established redistribution permission for every separate RewardBench result or contributed WMT score; benchmark and toolkit licenses do not resolve the status of every asset. The data archive remains local. Intended use is research on retention certification. Generative AI tools supported problem framing, proof discussion, coding, analysis and manuscript drafting. Human authors remain responsible for claims, code, results, citations and final text. Historical compute accounting is incomplete; `RESPONSIBLE_NLP_CURRENT.md` records available details.

References

Fariya Afrin and Ibne Farabi Shihab. 2026. Evaluator ensembles under reward hacking: Covariance geometry and finite-search guarantees. Preprint, arXiv:2608.08002.

Antoine Amarilli, Yael Amsterdamer, Tova Milo, and Pierre Senellart. 2017. Top-k querying of unknown values under order constraints. In *20th International Conference on Database Theory (ICDT 2017)*, volume 68 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 5:1–5:18. Schloss Dagstuhl – Leibniz-Zentrum für Informatik.

Anastasios N. Angelopoulos, Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster. 2024. Conformal risk control. In *The Twelfth International Conference on Learning Representations*, Vienna, Austria. OpenReview.net.

Abolfazl Asudeh, H. V. Jagadish, Gerome Miklau, and Julia Stoyanovich. 2018. On obtaining stable rankings. *Proceedings of the VLDB Endowment*, 12(3):237–250.

Sher Badshah, Ali Emami, and Hassan Sajjad. 2026. SCOPE: Selective conformal optimized pairwise LLM judging. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306, pages 4894–4916. PMLR.

Tian Bai and Ying Jin. 2026. Conformal selective prediction with general risk control. Preprint, arXiv:2603.24704.

Stephen Bates, Anastasios N. Angelopoulos, Lihua Lei, Jitendra Malik, and Michael I. Jordan. 2021. Distribution-free, risk-controlling prediction sets. *Journal of the ACM*, 68(6):43:1–43:34.

Dimitris Bertsimas and Melvyn Sim. 2004. The price of robustness. *Operations Research*, 52(1):35–53.

Emily Black, Klas Leino, and Matt Fredrikson. 2022. Selective ensembles for consistent predictions. In *The Tenth International Conference on Learning Representations*. OpenReview.net.

Guy Blanc, Caleb Koch, Jane Lange, and Li-Yang Tan. 2022. The query complexity of certification. In *Proceedings of the 54th Annual ACM SIGACT Symposium on Theory of Computing*, STOC '22, pages 623–636. ACM.

Francesco Busolin, Claudio Lucchese, Franco Maria Nardini, Salvatore Orlando, Raffaele Perego, and Salvatore Trani. 2021. Learning early exit strategies for additive ranking ensembles. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2217–2221. Association for Computing Machinery.

Moses Charikar, Ronald Fagin, Venkatesan Guruswami, Jon Kleinberg, Prabhakar Raghavan, and Amit Sahai. 2002. Query strategies for priced information. *Journal of Computer and System Sciences*, 64(4):785–819.

André Chassein, Marc Goerigk, Adam Kasperski, and Paweł Zieliński. 2018. On recoverable and two-stage robust selection problems with budgeted uncertainty. *European Journal of Operational Research*, 265(2):423–436.

Xiaoyu Chen, Marc Goerigk, and Michael Poss. 2026. The robust selection problem with information discovery. *Discrete Applied Mathematics*, 382:277–292.

Ming Cheng, Yusheng Dai, Qiuhong Ke, Zhaolin Chen, and Lizhen Qu. 2026. Multi-expert conformal risk control for pairwise LLM judging in open-ended dialogue. Preprint, arXiv:2608.26529.

Thomas Coste, Usman Anwar, Robert Kirk, and David Krueger. 2024. Reward model ensembles help mitigate overoptimization. In *The Twelfth International Conference on Learning Representations*. OpenReview.net.

Amol Deshpande, Lisa Hellerstein, and Devorah Kletenik. 2016. Approximation algorithms for stochastic submodular set cover with applications to Boolean function evaluation and min-knapsack. *ACM Transactions on Algorithms*, 12(3):42:1–42:28.

Daniel Deutsch, George Foster, and Markus Freitag. 2023. Ties matter: Meta-evaluating modern metrics with pairwise accuracy and tie calibration. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12914–12929, Singapore. Association for Computational Linguistics.

Haya Diwan, Lisa Hellerstein, Nicole Megow, and Jens Schlöter. 2026. Optimal verification of a minimum-weight basis in an uncertainty matroid. In *43rd International Symposium on Theoretical Aspects of Computer Science (STACS 2026)*, volume 364 of *Leibniz International Proceedings in Informatics*, pages 32:1–32:20. Schloss Dagstuhl – Leibniz-Zentrum für Informatik.

Jacob Eisenstein, Chirag Nagpal, Alekh Agarwal, Ahmad Beirami, Alexander D'Amour, DJ Dvijotham, Adam Fisch, Katherine Heller, Stephen Pfohl, Deepak Ramachandran, Peter Shaw, and Jonathan Berant. 2024. Helping or herding? reward model ensembles mitigate but do not eliminate reward hacking. In *First Conference on Language Modeling*. OpenReview.net.

Thomas Erlebach and Michael Hoffmann. 2015. Query-competitive algorithms for computing with uncertainty. *Bulletin of the EATCS*, 116:22–39.

Thomas Erlebach, Michael Hoffmann, and Murilo Santos de Lima. 2021. Round-competitive algorithms for uncertainty problems with parallel queries. In *38th International Symposium on Theoretical Aspects of Computer Science (STACS 2021)*, volume 187 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 27:1–27:18. Schloss Dagstuhl – Leibniz-Zentrum für Informatik.

Alexander R. Fabbri, Wojciech Kryściński, Bryan McCann, Caiming Xiong, Richard Socher, and Dragomir Radev. 2021. SummEval: Re-evaluating summarization evaluation. *Transactions of the Association for Computational Linguistics*, 9:391–409.

Ronald Fagin, Amnon Lotem, and Moni Naor. 2003. Optimal aggregation algorithms for middleware. *Journal of Computer and System Sciences*, 66(4):614–656.

Markus Freitag, Nitika Mathur, Daniel Deutsch, Chi-Kiu Lo, Eleftherios Avramidis, Ricardo Rei, Brian Thompson, Frederic Blain, Tom Kocmi, Jiayi Wang, David Ifeoluwa Adelani, Marianna Buchicchio, Chrysoula Zerva, and Alon Lavie. 2024. Are LLMs breaking MT metrics? results of the WMT24 metrics shared task. In *Proceedings of the Ninth Conference on Machine Translation*, pages 47–81, Miami, Florida, USA. Association for Computational Linguistics.

Markus Freitag, Nitika Mathur, Chi-kiu Lo, Eleftherios Avramidis, Ricardo Rei, Brian Thompson, Tom Kocmi, Frederic Blain, Daniel Deutsch, Craig Stewart, Chrysoula Zerva, Sheila Castilho, Alon Lavie,

and George Foster. 2023. Results of WMT23 metrics shared task: Metrics might be guilty but references are not innocent. In *Proceedings of the Eighth Conference on Machine Translation*, pages 578–628, Singapore. Association for Computational Linguistics.

Dimitrios Gkenosis, Nathaniel Grammel, Lisa Hellerstein, and Devorah Kletenik. 2018. The stochastic score classification problem. In *26th Annual European Symposium on Algorithms (ESA 2018)*, volume 112 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 36:1–36:14, Dagstuhl, Germany. Schloss Dagstuhl – Leibniz-Zentrum für Informatik.

Marc Goerigk, Manoj Gupta, Jonas Ide, Anita Schöbel, and Sandeep Sen. 2015. The robust knapsack problem with queries. *Computers & Operations Research*, 55:12–22.

Meghal Gupta and Naren Sarayu Manoj. 2023. An optimal algorithm for certifying monotone functions. In *Proceedings of the 2023 Symposium on Simplicity in Algorithms (SOSA)*, pages 207–212. Society for Industrial and Applied Mathematics.

Aviram Imber and Benny Kimelfeld. 2020. The complexity of determining the necessary and possible top-k winners in partial voting profiles. Preprint, arXiv:2002.09212.

Tomasz Imieliński and Witold Lipski, Jr. 1984. Incomplete information in relational databases. *Journal of the ACM*, 31(4):761–791.

Hiroshi Inoue. 2019. Adaptive ensemble prediction for deep neural networks based on confidence level. In *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics*, volume 89 of *Proceedings of Machine Learning Research*, pages 1284–1293. PMLR.

Shivaram Kalyanakrishnan, Ambuj Tewari, Peter Auer, and Peter Stone. 2012. PAC subset selection in stochastic multi-armed bandits. In *Proceedings of the 29th International Conference on Machine Learning (ICML)*, pages 655–662, New York, NY, USA. Omnipress.

Bum Jun Kim. 2026. Lottery tickets are not deployment tickets. Preprint, arXiv:2607.27031.

Minghan Li, Xinyu Zhang, Ji Xin, Hongyang Zhang, and Jimmy Lin. 2022. Certified error control of candidate set pruning for two-stage relevance ranking. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 333–345, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Thodoris Lykouris and Sergei Vassilvitskii. 2018. Competitive caching with machine learned advice. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 3296–3305. PMLR.

Saumya Malik, Valentina Pyatkin, Sander Land, Jacob Morrison, Noah A. Smith, Hannaneh Hajishirzi, and Nathan Lambert. 2026. RewardBench 2: Advancing reward model evaluation. In *The Fourteenth International Conference on Learning Representations*, Rio de Janeiro, Brazil. OpenReview.net.

Nicole Megow and Jens Schlöter. 2023. Set selection under explorable stochastic uncertainty via covering techniques. In *Integer Programming and Combinatorial Optimization*, pages 319–333. Springer International Publishing.

Mahdi Milani Fard, Quentin Cormier, Kevin Canini, and Maya Gupta. 2016. Launch and iterate: Reducing prediction churn. In *Advances in Neural Information Processing Systems*, volume 29.

Lutz Oettershagen. 2026. Top-k on a budget: Adaptive ranking with weak and strong oracles. Preprint, arXiv:2601.20989.

Marios Papamichalis, Regina Ruane, and Theofanis Papamichalis. 2026. Replicable conformal prediction. Preprint, arXiv:2608.23638.

Stefano Perrella, Lorenzo Proietti, Pere-Lluís Huguet Cabot, Edoardo Barba, and Roberto Navigli. 2024. Beyond correlation: Interpretable evaluation of machine translation metrics. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 20689–20714, Miami, Florida, USA. Association for Computational Linguistics.

Roi Pony, Adi Raz Goldfarb, Oshri Naparstek, Idan Friedman, Udi Barzelay, and Eli Schwartz. 2026. Col-bandit: Query-time top-k estimation for late-interaction retrieval. Preprint, arXiv:2602.02827.

Gordan Prastalo and Kevin Maik Jablonka. 2026. Reducing cross-sample prediction churn in scientific machine learning. Preprint, arXiv:2605.13826.

Manish Purohit, Zoya Svitkina, and Ravi Kumar. 2018. Improving online algorithms via ML predictions. In *Advances in Neural Information Processing Systems*, volume 31.

Jens Schlöter. 2025. On the complexity of knapsack under explorable uncertainty: Hardness and algorithms. In *33rd Annual European Symposium on Algorithms (ESA 2025)*, volume 351 of *Leibniz International Proceedings in Informatics*, pages 6:1–6:15. Schloss Dagstuhl – Leibniz-Zentrum für Informatik.

Behzad Shayegh, Jan-Thorsten Peter, David Vilar, Tobias Domhan, Juraj Juraska, Markus Freitag, and Lili Mou. 2025. Feeding two birds or favoring one? adequacy–fluency tradeoffs in evaluation and meta-evaluation of machine translation. In *Proceedings of the Tenth Conference on Machine Translation*, pages 269–285, Suzhou, China. Association for Computational Linguistics.

Mohamed A. Soliman, Ihab F. Ilyas, and Kevin Chen-Chuan Chang. 2007. Top-k query processing in uncertain databases. In *Proceedings of the 23rd International Conference on Data Engineering (ICDE)*, pages 896–905. IEEE.

Aman Singh Thakur, Kartik Choudhary, Venkat Srinik Ramayapally, Sankaran Vaidyanathan, and Dieuwke Hupkes. 2024. Judging the judges: Evaluating alignment and vulnerabilities in LLMs-as-judges. Preprint, arXiv:2406.12624.

Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghuai Lin, Yunbo Cao, Lingpeng Kong, Qi Liu, Tianyu Liu, and Zhifang Sui. 2024. Large language models are not fair evaluators. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9440–9450, Bangkok, Thailand. Association for Computational Linguistics.

Jamelle Watson-Daniels, Flavio du Pin Calmon, Alexander D'Amour, Carol Long, David C. Parkes, and Berk Ustun. 2025. Predictive churn with the set of good models. Preprint, arXiv:2402.07745.

Chen Wu, Ruqing Zhang, Jiafeng Guo, Wei Chen, Yixing Fan, Maarten de Rijke, and Xueqi Cheng. 2022. Certified robustness to word substitution ranking attack for neural ranking models. In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*, pages 2128–2137. Association for Computing Machinery.

Christopher Yeh, Nicolas Christianson, Adam Wierman, and Yisong Yue. 2025. Conformal risk training: End-to-end optimization of conformal risk control. In *Advances in Neural Information Processing Systems*, volume 38, pages 70056–70092. Curran Associates, Inc.

Cenwei Zhang, Teng Fang, Yuxia Wang, Derek Li, Bryan Dai, and Lei You. 2026. Best-of-evidence: Best-of-N selection under partial verification. Preprint, arXiv:2607.20950.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems*, volume 36, pages 46595–46623, New Orleans, LA, USA. Curran Associates, Inc.